

FEE3411: Control Systems A
Week 4
Block Diagrams and Signal Flow Graphs

What this week covers

By the end of this week you should be able to

- Draw the block diagram of an interconnected system, using the three elements — block, summing junction and pick-off point — and state the no-loading assumption that the cascade rule depends on.
- Reduce the three basic connections (cascade, parallel, feedback) to a single block, deriving the closed-loop formula rather than quoting it.
- Identify, for a system in canonical feedback form, the forward-path, open-loop, closed-loop and error transfer functions, and write down the characteristic equation.
- Apply the rules of block-diagram algebra — including moving a summing junction or a pick-off point past a block — to reduce a multi-loop diagram to one transfer function.
- Convert a block diagram to a signal flow graph, and identify its forward paths, loops, loop gains and non-touching loops.
- Evaluate a transfer function with Mason's gain formula, including the non-touching-loop terms of Δ and the cofactors Δ_k , and check the result against block-diagram reduction.
- Find the response of a system with more than one input by superposition, computing one transfer function per input.

Reading

Primary — Khalil, §2-3 Block Diagrams (pp. 40–49).

Secondary — Nise (6th ed.), §5.2 Block Diagrams (p. 236); §5.3 Analysis and Design of Feedback Systems (p. 245); §5.4 Signal-Flow Graphs (p. 248); §5.5 Mason's Rule (p. 251).
Khalil does not cover signal flow graphs — for §6–§7 Nise is the primary text.

Further — Nagrath & Gopal (2nd ed.), §2.4 Block Diagram Algebra (p. 38), §2.5 Signal Flow Graphs (p. 45); Sundararajan, §5.1, §5.2, §5.2.1.

Contact hours this week: 2 h lecture + 1 h tutorial.

Syllabus items addressed: “Block diagrams and canonical feedback. Derivation of transfer functions from the reduction of elementary block diagrams. Construction of signal flow diagrams. Block diagram reduction using signal flow diagrams.” (Transfer Functions); “Block diagrams” (Time Domain Linear Systems Analysis).

Expected learning outcomes: ELO 2 — *analyse a control system solution in the time- and frequency-domains*; ELO 1 — *describe the working principles of a given control system* (the block diagram is how a control system's working principle is described on paper).

A note on the two halves. The first hour is block diagrams: the elements, the three basic connections, the canonical feedback form, and the algebra for reducing a multi-loop diagram by

hand. That algebra works, but it gets awkward as soon as two loops overlap, and §5 shows exactly where it starts to hurt. The second hour answers that with signal flow graphs and Mason's gain formula, which read the same answer off the same diagram without redrawing it once. Week 3 gave us the transfer function of a single component; this week is about what happens when components are wired together, which is the last piece of machinery needed before Week 5 starts asking how these systems actually respond.

Part A — Block diagrams

1 From one block to many

Week 3 produced transfer functions for single components: a network, a motor, a mass on a spring. A real control system is several of these wired together — a reference input compared against a measurement, the difference amplified, the amplifier driving a motor, the motor's shaft position measured by a sensor that closes the loop. To get the transfer function of the whole thing we need a notation for the wiring, and rules for collapsing it.

A **block diagram** is that notation. It has exactly three elements, shown in Figure 1.

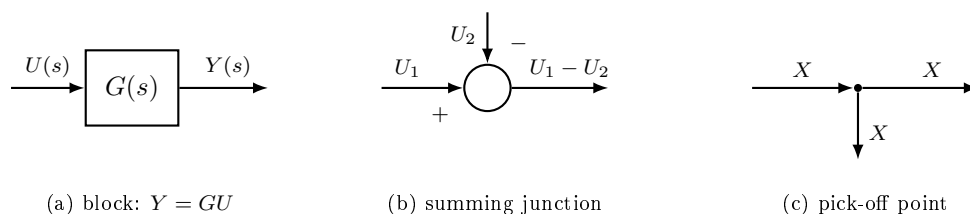


Figure 1: The three elements of a block diagram. A block multiplies its input by a transfer function; a summing junction adds signals with the signs marked beside it; a pick-off point sends the *same* signal down two branches. Cf. Khalil Figs. 2-31 and 2-32, pp. 40–41.

Two things about this notation matter more than they first appear.

First, a **pick-off point does not divide the signal**. Both branches carry X in full. This is the opposite of a wire junction in a circuit, where current divides; the block diagram is a diagram of *signals*, not of power flow, and taking a copy of a signal costs nothing.

Second, and more consequentially: writing two blocks in cascade and multiplying their transfer functions assumes the second does not *load* the first.

Common pitfall

Cascading G_1 and G_2 to get G_1G_2 assumes that connecting G_2 does not change G_1 's output — that is, that G_2 draws no current (or torque, or flow) from G_1 . Two RC networks in series are the standard trap: each alone has transfer function $1/(RCs + 1)$, but wired directly together their combined transfer function is *not* $1/(RCs + 1)^2$, because the second network's input impedance loads the first. Cascade blocks only where a buffer amplifier separates the stages, or where the impedances make loading negligible — and if a question gives you a circuit rather than a diagram, go back to the impedance method of Week 3 and derive the whole thing at once.

Key idea

A block diagram is a picture of a set of simultaneous algebraic equations in s — one equation per block and per summing junction. Every reduction rule in this week's work is nothing more than eliminating an internal variable between those equations. If a diagram ever defeats you, write its equations out and solve them; you will always get the right answer, just more slowly.

2 The three basic connections

Every block diagram, however tangled, is built from three connections: cascade, parallel and feedback. Figure 2 shows each with its single-block equivalent.

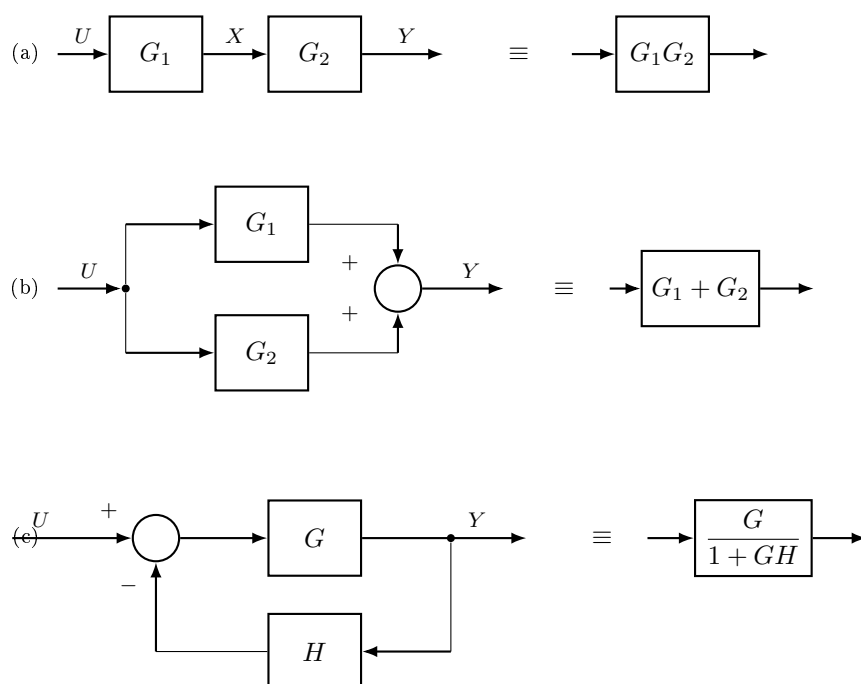


Figure 2: The three basic connections and their single-block equivalents: (a) cascade, (b) parallel, (c) negative feedback. Cf. Khalil Fig. 2-34, p. 41; Nise §5.2, pp. 236–240.

Cascade. With $X = G_1U$ and $Y = G_2X$, substitution gives $Y = G_2G_1U$, so the equivalent transfer function is the *product* G_1G_2 . Order does not matter algebraically, because transfer functions are scalar rational functions and multiplication commutes — which is worth noticing, because the physical systems certainly do not commute. Swapping a real amplifier and a real motor would matter; swapping their transfer functions does not.

Parallel. With both blocks fed the same U from a pick-off point and their outputs summed, $Y = G_1U \pm G_2U$, so the equivalent is $G_1 \pm G_2$ — a *sum*, with whatever signs the summing junction carries.

Feedback. This one deserves the full derivation, because it is the formula the rest of the unit runs on. Let the summing junction subtract the feedback signal, so that with E the signal at the junction's output,

$$E = U - HY, \quad Y = GE.$$

Eliminating E between the two,

$$Y = G(U - HY) \implies Y(1 + GH) = GU \implies \boxed{\frac{Y}{U} = \frac{G}{1 + GH}}.$$

If the junction *adds* the feedback signal instead — positive feedback — the same two lines give $G/(1 - GH)$. The sign in the denominator is therefore the opposite of the sign at the summing junction, which is the single most common place to lose a minus sign in this entire unit.

Common pitfall

Read the sign off the *diagram*, not off memory. The formula $G/(1 + GH)$ is for negative feedback, where the junction subtracts. A diagram with a $+$ at the feedback input needs $G/(1 - GH)$, and that system is a completely different animal: its denominator can vanish, which is exactly what instability looks like. When a question's answer comes out with a suspicious sign, check the summing junction first.

3 The canonical feedback form

Almost every control system in this unit is drawn, eventually, in the form of Figure 3: a reference in, a comparison, a forward path, a measurement path back. It is worth having names for its parts, because examination questions and textbooks both use them freely.

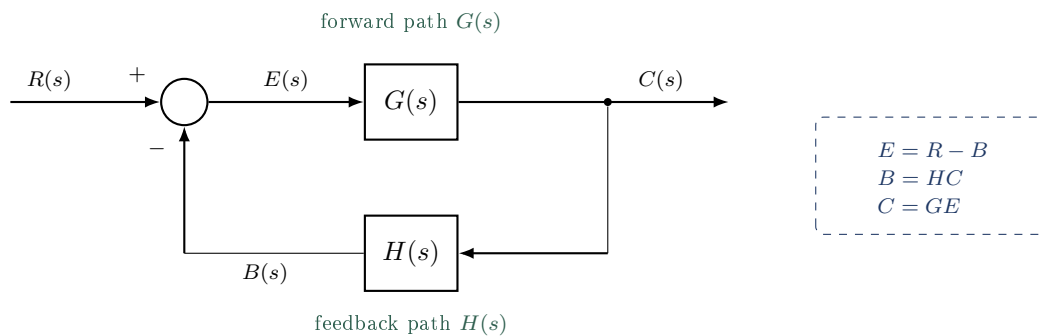


Figure 3: The canonical negative-feedback form, with the three equations it encodes. R is the reference, C the controlled output, B the feedback signal and E the actuating error signal. Cf. Nagrath & Gopal Fig. 2.20, p. 40.

Eliminating E and B from the three equations beside the figure gives the four transfer functions that get quoted by name:

$$\begin{aligned} \text{forward-path (or plant) transfer function: } & G(s) \\ \text{open-loop transfer function (loop gain): } & G(s)H(s) \\ \text{closed-loop transfer function: } & T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \quad (1) \\ \text{error transfer function: } & \frac{E(s)}{R(s)} = \frac{1}{1 + G(s)H(s)} \quad (2) \end{aligned}$$

When $H(s) = 1$ the system is called **unity feedback**, the measurement is taken as perfect, and E becomes the true error $R - C$.

Key idea

The denominator of the closed-loop transfer function is $1 + G(s)H(s)$, so the closed-loop poles are the roots of

$$1 + G(s)H(s) = 0,$$

the **characteristic equation** of the closed-loop system. This single equation is the object of Weeks 7, 8, 11 and 12: Routh–Hurwitz tests whether its roots are in the left half plane, the root locus tracks where they move as a gain varies, and the Nyquist criterion counts them from a frequency-response plot. Everything downstream of this week is about the roots of $1 + GH = 0$.

Common pitfall

The open-loop transfer function is $G(s)H(s)$, the gain around the whole loop — not the forward path $G(s)$ alone, and not the closed-loop $T(s)$ with the loop cut. The three are equal only in the trivial case $H = 1$ and no feedback at all. When a question says “for the open-loop transfer function $G(s)H(s) = K/[s(s+2)]$ ”, it is handing you the *product*, and the characteristic equation is $1 + K/[s(s+2)] = 0$ directly.

Worked example 4.1 — what feedback does to the poles

A unity-feedback system has forward path

$$G(s) = \frac{8}{s(s+6)}, \quad H(s) = 1.$$

Find the closed-loop transfer function, its poles, its d.c. gain, and the error transfer function. Compare the closed-loop poles with the open-loop poles.

Closed-loop transfer function. From (1),

$$T(s) = \frac{G}{1 + GH} = \frac{\frac{8}{s(s+6)}}{1 + \frac{8}{s(s+6)}} = \frac{8}{s(s+6) + 8} = \frac{8}{s^2 + 6s + 8} = \frac{8}{(s+2)(s+4)}.$$

The middle step — multiplying numerator and denominator by $s(s+6)$ — is the one worth doing carefully; it is where the 1 in $1 + GH$ turns into the open-loop denominator.

Poles. Open loop, G has poles at $s = 0$ and $s = -6$: one of them is at the origin, so the open-loop system is only marginally stable and its step response would ramp away without bound. Closed loop, the poles are at $s = -2$ and $s = -4$ — both strictly in the left half plane. Closing the loop has moved both poles and made the system stable.

d.c. gain. $T(0) = 8/8 = 1$. The closed-loop system follows a constant reference exactly, which is what the free integrator in G buys us — a result Week 6 will make precise as “a type-1 system has zero steady-state error to a step”.

Error transfer function. From (2),

$$\frac{E(s)}{R(s)} = \frac{1}{1 + GH} = \frac{s(s+6)}{s^2 + 6s + 8},$$

whose d.c. gain is 0 — the same statement as $T(0) = 1$, read from the other side.

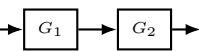
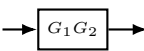
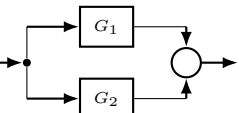
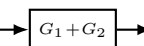
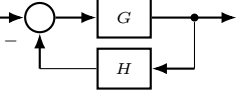
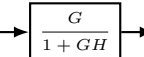
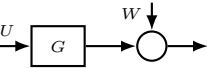
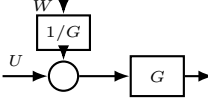
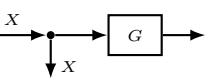
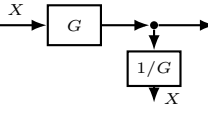
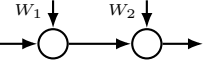
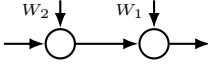
What it means. Feedback did not add or remove any modes: the closed-loop system still has two poles, because $1 + GH = 0$ still has degree two. What it did was *relocate* them,

from $\{0, -6\}$ to $\{-2, -4\}$. That relocation is the whole point of feedback, and choosing where to put the poles is what FEE3412 calls design.

4 Block-diagram algebra

The three basic connections handle diagrams whose loops are neatly nested. Real diagrams need a few more moves, collected in Table 1. Each is proved the same way: write the equations for the signals before and after, and check they agree.

Table 1: The rules of block-diagram algebra. Each row's two diagrams are equivalent. Cf. Nagrath & Gopal Table 2.4, p. 42; Nise §5.2, p. 236.

Rule	Original	Equivalent
Combine blocks in cascade		
Combine blocks in parallel		
Eliminate a feedback loop		
Move a summing junction back through a block		
Move a pick-off point forward through a block		
Interchange adjacent summing junctions		

The two *moving* rules are the ones that need thought, and both follow the same logic: a signal must be worth the same after the move as before.

Moving a summing junction. In the original, the added signal W enters *after* the block, so the output is $GU + W$. To move the junction to the block's input and keep the output the same, whatever we add there gets multiplied by G on the way through — so we must add W/G , giving $G(U + W/G) = GU + W$ as required. Moving a junction *against* the signal flow divides by the block; moving it *with* the flow multiplies by it.

Moving a pick-off point. In the original, the branch taps X before the block. Moved to after the block, the branch would carry GX , which is wrong by a factor G — so the moved branch needs a $1/G$ block to undo it. Same principle, opposite direction: the rule always *compensates* for what the block does to the signal on the moved path.

Common pitfall

A $1/G$ block introduced by one of these moves is a bookkeeping device, not a physical component — and it is not necessarily a proper transfer function. If $G = 1/(s + 2)$ then $1/G = s + 2$, a pure differentiator plus a gain, which no hardware realises exactly. That is fine: the $1/G$ block will cancel before the reduction finishes. It is only a problem if you stop halfway and try to interpret the intermediate diagram as a real system.

Worked example 4.2 — a servo with tachometer feedback

A position servo consists of an amplifier of gain $G_1 = 4$ driving a motor with transfer function

$$G_2(s) = \frac{1}{s(s+2)} \quad (\text{shaft angle per volt}),$$

which is the reduced servomotor model of Week 2. A tachometer on the motor shaft measures shaft *speed* and feeds it back to the amplifier's input through a junction; since speed is s times position, the tachometer path has transfer function $H_1(s) = s$. The shaft position is then measured by a potentiometer of unit gain and fed back to the outer summing junction. Find $C(s)/R(s)$, and identify ω_n and ζ of the result.

Step 1 — collapse the inner loop. The inner loop is a plain negative-feedback connection of G_2 with H_1 :

$$\frac{G_2}{1 + G_2 H_1} = \frac{\frac{1}{s(s+2)}}{1 + \frac{s}{s(s+2)}} = \frac{1}{s(s+2) + s} = \frac{1}{s^2 + 3s} = \frac{1}{s(s+3)}.$$

Look carefully at what happened: the motor's pole at $s = -2$ has moved to $s = -3$. Tachometer feedback has made the motor faster — or, in the language of Week 5, it has added damping. This is why tachometer feedback is fitted to real servos, and it is the reason the inner loop exists at all.

Step 2 — cascade the amplifier. The forward path of the remaining single loop is

$$G_1 \cdot \frac{1}{s(s+3)} = \frac{4}{s(s+3)}.$$

Step 3 — collapse the outer loop (unity feedback):

$$T(s) = \frac{\frac{4}{s(s+3)}}{1 + \frac{4}{s(s+3)}} = \frac{4}{s(s+3) + 4} = \frac{4}{s^2 + 3s + 4}.$$

Reading the answer. Comparing $s^2 + 3s + 4$ with the standard second-order denominator $s^2 + 2\zeta\omega_n s + \omega_n^2$,

$$\omega_n = \sqrt{4} = 2 \text{ rad s}^{-1}, \quad 2\zeta\omega_n = 3 \implies \zeta = \frac{3}{4} = 0.75.$$

The poles are at $s = -1.5 \pm j1.32$: a stable, underdamped system with d.c. gain $T(0) = 4/4 = 1$. Week 5 turns $\omega_n = 2$ and $\zeta = 0.75$ directly into an overshoot and a settling time; this week's job is only to get the transfer function they are read from.

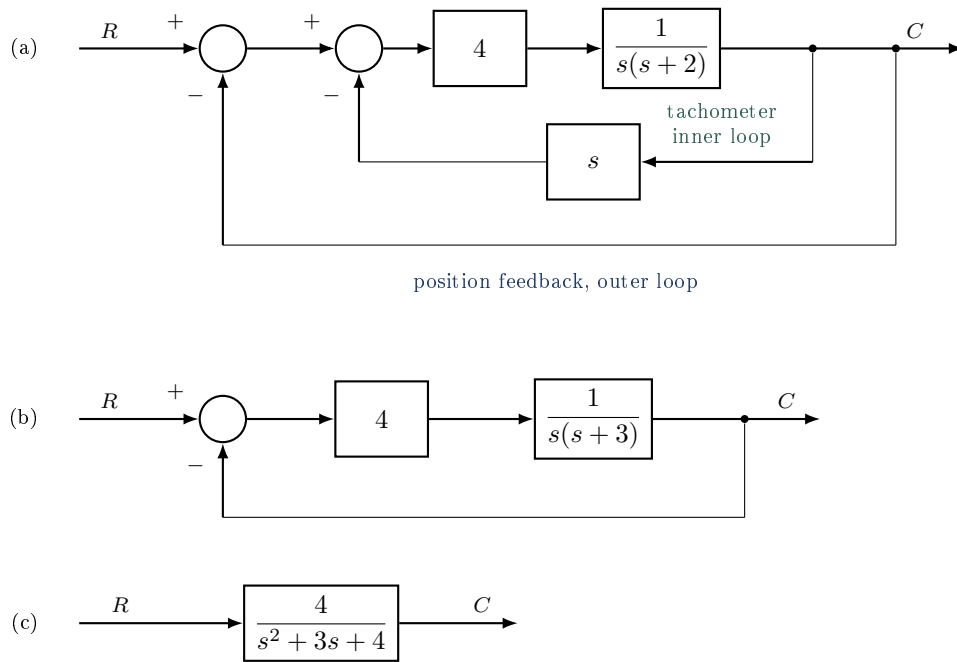


Figure 4: Worked example 4.2 reduced in stages: (a) the servo with tachometer and position feedback, (b) after the inner loop is collapsed — note the motor pole moving from -2 to -3 , (c) the single closed-loop block.

5 When reduction gets awkward: overlapping loops

Worked example 4.2 was easy because its loops were **nested**: the inner loop sat entirely inside the outer one, so collapsing it left a diagram of the same shape. Loops can also **overlap** — each containing part, but not all, of the other — and then no amount of collapsing helps, because there is no innermost loop to start from. Figure 5(a) is the smallest example, and it is the standard one in every textbook.

Worked example 4.3 — overlapping loops by block-diagram algebra

For Figure 5(a), with

$$G_1 = 5, \quad G_2 = \frac{1}{s+2}, \quad G_3 = \frac{1}{s+1}, \quad H_1 = 1, \quad H_2 = 8,$$

find $T(s) = C(s)/R(s)$.

Step 1 — move the pick-off point. Loop 1 is fed from the output of G_2 . Move that pick-off forward through G_3 , compensating the branch with $1/G_3 = s+1$, as in Figure 5(b). Loop 1 now taps the same node as loop 2, and loop 2 sits inside it: the loops are nested.

Step 2 — collapse the inner loop (G_2G_3 with H_2):

$$\frac{G_2G_3}{1 + G_2G_3H_2} = \frac{1}{(s+1)(s+2)} = \frac{1}{(s+1)(s+2) + 8} = \frac{1}{s^2 + 3s + 10}.$$

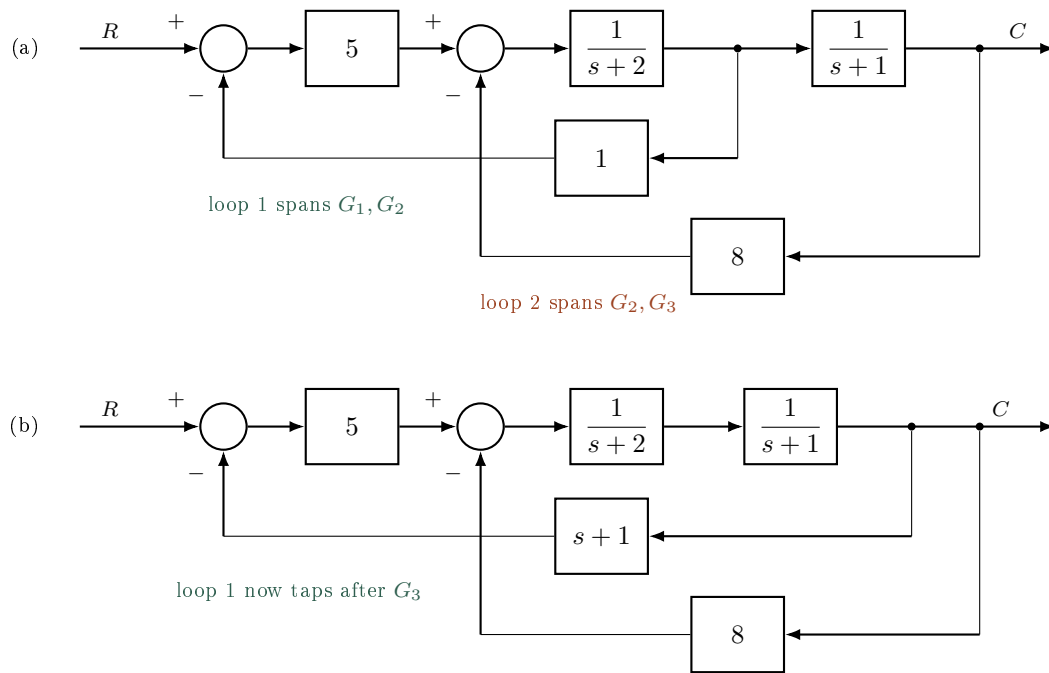


Figure 5: Overlapping loops. In (a) neither loop contains the other, so there is no inner loop to collapse first. In (b) the pick-off point of loop 1 has been moved forward through G_3 — its gain picking up a factor $s + 1$ in compensation — which makes the two loops nested and the reduction routine.

Step 3 — collapse the outer loop. Its forward path is G_1 times the result of step 2, and its feedback path is $H_1(s + 1) = s + 1$:

$$T(s) = \frac{5}{s^2 + 3s + 10 + \frac{5(s+1)}{s^2 + 3s + 10}} = \frac{5}{s^2 + 3s + 10 + 5s + 5} = \frac{5}{s^2 + 8s + 15} = \frac{5}{(s+3)(s+5)}.$$

Check. The d.c. gain is $T(0) = 5/15 = 1/3$, and the closed-loop poles are at -3 and -5 : real, distinct and stable. Notice that the two open-loop poles were at -1 and -2 ; feedback has again relocated them, this time further into the left half plane.

The cost. Getting here took a junction move, two loop collapses and a redrawn diagram, and every one of those steps is a place to lose a factor. The answer is right, but the method does not scale — a diagram with three overlapping loops would need several moves, and each redrawing invites error. That is the problem §7 solves.

Part B — Signal flow graphs and Mason's gain formula

6 Signal flow graphs

A **signal flow graph** (SFG) carries exactly the same information as a block diagram in a leaner notation, invented by S. J. Mason precisely so that the reduction could be replaced by a formula.

- A **node** is a signal. It is drawn as a small dot and labelled with the signal's name.
- A **branch** is a directed line from one node to another, labelled with its gain. Travelling along it multiplies the signal by that gain.
- The signal at any node is the *sum* of everything arriving on the branches that end there.

That last rule is what removes the need for summing junctions: addition is built into the meaning of a node, and a subtraction is carried by a branch of gain $-H$ rather than a $-$ sign on a circle. Table 2 is the translation, and Figure 6 shows it applied to the canonical feedback form.

Table 2: Block diagram and signal flow graph: the same content, two notations.

Block diagram	Signal flow graph	Note
Block with transfer function G	Branch of gain G	The gain sits on the line, not in a box
Signal on a connecting line	Node	Named signals become dots
Summing junction	Node (addition is implied)	Signs move onto the branch gains
Pick-off point	Node with two or more outgoing branches	Each branch carries the node's full value
Input (reference)	Source node — only outgoing branches	Also called an input node
Output (controlled variable)	Sink node — only incoming branches	Add a unit branch if the output node also feeds a loop

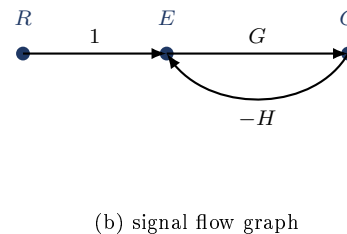
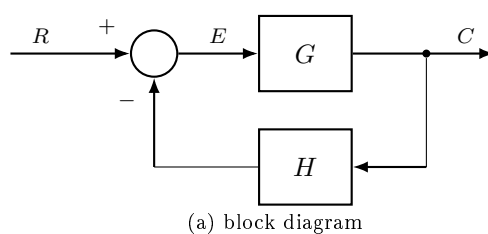


Figure 6: The canonical feedback form in both notations. The summing junction becomes the node E , and its minus sign is absorbed into the feedback branch gain $-H$. Cf. Nise §5.4, p. 248; Nagrath & Gopal Fig. 2.25, p. 45.

The vocabulary Mason's formula needs is all about routes through the graph. Figure 7 is a graph carrying one of each.

- A **path** is any route along branches, following the arrows, that passes through no node twice.
- A **forward path** is a path from the source node to the sink node. Its **forward-path gain** is the product of the gains along it.
- A **loop** is a closed path that starts and ends at the same node, visiting no other node twice. Its **loop gain** is the product of the gains around it — including the minus signs.
- Two loops are **non-touching** if they share *no node at all*.

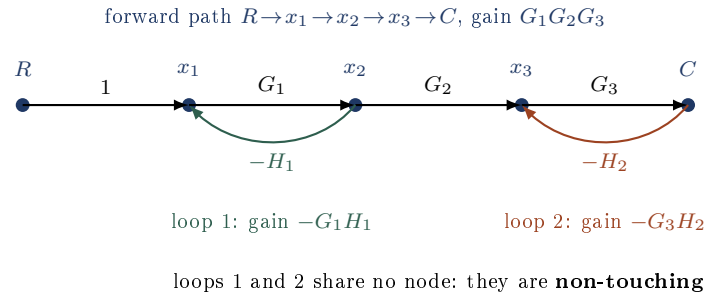


Figure 7: Signal flow graph vocabulary. Loop 1 touches nodes x_1, x_2 and loop 2 touches x_3, C ; because those sets are disjoint, the two loops are non-touching, and Mason’s formula will carry a product term for the pair.

Common pitfall

“Non-touching” means *no shared node* — not “drawn without crossing”, and not “no shared branch”. Two loops that meet at a single node are touching, and contribute no product term. Getting this wrong is the most common source of a wrong Δ , and because the error is in a term you simply failed to write, nothing in the algebra flags it. Check every pair of loops explicitly: list each loop’s nodes, then compare the lists.

7 Mason’s gain formula

Mason’s gain formula reads the transfer function between a source node and a sink node straight off the graph, with no redrawing at all:

$$T(s) = \frac{\sum_k P_k \Delta_k}{\Delta}, \quad (3)$$

where

- P_k is the gain of the k -th forward path, summed over *all* forward paths;
- Δ is the **determinant** of the graph, built from the loop gains as in (4) below;
- Δ_k is the **cofactor** of path k : the determinant Δ recomputed with every loop that *touches* path k deleted from the graph.

The determinant is an alternating sum over the loops, taken one at a time, then in non-touching pairs, then in non-touching triples:

$$\Delta = 1 - \sum_i L_i + \sum_{\substack{\text{pairs } i,j \\ \text{non-touching}}} L_i L_j - \sum_{\substack{\text{triples } i,j,k \\ \text{non-touching}}} L_i L_j L_k + \cdots \quad (4)$$

where L_i is the gain of the i -th loop, signs included. In this unit the sum almost always stops at the pairs; triples need three loops that mutually share no node, which takes a larger graph than any examination question here will set.

Two special cases are worth noting before we use it. If every loop touches every forward path — which is true of most single-input single-output control diagrams — then every $\Delta_k = 1$ and (3) collapses to $(\sum_k P_k) / \Delta$. And for the canonical feedback form of Figure 6, there is one forward path $P_1 = G$ and one loop $-GH$, so $\Delta = 1 + GH$, $\Delta_1 = 1$, and Mason gives $T = G / (1 + GH)$ — the formula of §2, recovered as the simplest possible case.

Key idea

Block-diagram reduction and Mason's formula are two routes to the same transfer function, and the choice between them is practical, not mathematical. Reduction is clearer for a diagram with two or three nested loops and tells a story about the physics as you go. Mason wins the moment loops overlap or forward paths multiply, because it never requires the diagram to be redrawn — and a diagram that is never redrawn cannot be redrawn wrongly.

Worked example 4.4 — the overlapping-loop system again, in one line

Take the system of Worked example 4.3 — Figure 5(a), with $G_1 = 5$, $G_2 = 1/(s+2)$, $G_3 = 1/(s+1)$, $H_1 = 1$, $H_2 = 8$ — and apply Mason's formula to it directly.

Forward paths. There is one:

$$P_1 = G_1 G_2 G_3 = \frac{5}{(s+1)(s+2)}.$$

Loops. There are two. Loop 1 runs forward through G_1 and G_2 and back through H_1 ; loop 2 runs forward through G_2 and G_3 and back through H_2 :

$$L_1 = -G_1 G_2 H_1 = -\frac{5}{s+2}, \quad L_2 = -G_2 G_3 H_2 = -\frac{8}{(s+1)(s+2)}.$$

They both pass through the node at the output of G_2 , so they *touch*: there is no non-touching pair, and Δ stops after the loop-gain sum.

Determinant and cofactor.

$$\Delta = 1 - (L_1 + L_2) = 1 + \frac{5}{s+2} + \frac{8}{(s+1)(s+2)}, \quad \Delta_1 = 1$$

($\Delta_1 = 1$ because the single forward path passes through both loops).

Result. Multiplying numerator and denominator of (3) by $(s+1)(s+2)$,

$$T(s) = \frac{P_1 \Delta_1}{\Delta} = \frac{5}{(s+1)(s+2) + 5(s+1) + 8} = \frac{5}{s^2 + 8s + 15} = \frac{5}{(s+3)(s+5)},$$

which is exactly the answer Worked example 4.3 obtained — with no pick-off move, no redrawn diagram and no intermediate transfer functions to mislay.

Worked example 4.5 — two forward paths and a non-touching pair

Figure 8 shows a graph with a feedforward branch and two loops that do not touch. Take

$$G_1 = 1, \quad G_2 = \frac{2}{s+1}, \quad G_3 = \frac{1}{s+2}, \quad G_4 = 4, \quad G_5 = 2, \quad H_1 = 3, \quad H_2 = 2.$$

Find $T(s) = C(s)/R(s)$.

Forward paths. Two:

$$P_1 = G_1 G_2 G_3 G_4 = \frac{8}{(s+1)(s+2)} \quad (\text{through the whole chain}),$$

$$P_2 = G_5 = 2 \quad (\text{the feedforward branch } x_1 \rightarrow C).$$

Loops. Two: $L_1 = -G_1H_1 = -3$ around x_1, x_2 , and $L_2 = -G_3H_2 = -2/(s+2)$ around x_3, x_4 . Their node sets $\{x_1, x_2\}$ and $\{x_3, x_4\}$ are disjoint, so this pair *is* non-touching and contributes a product term:

$$\Delta = 1 - (L_1 + L_2) + L_1L_2 = 1 + 3 + \frac{2}{s+2} + \frac{6}{s+2} = 4 + \frac{8}{s+2} = \frac{4(s+4)}{s+2}.$$

Cofactors. Path P_1 runs through every node of both loops, so $\Delta_1 = 1$. Path P_2 leaves the chain at x_1 , so it touches loop 1 but *not* loop 2 — loop 2 survives in its cofactor:

$$\Delta_2 = 1 - L_2 = 1 + \frac{2}{s+2} = \frac{s+4}{s+2}.$$

Result.

$$T(s) = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{\frac{8}{(s+1)(s+2)} + \frac{2(s+4)}{s+2}}{\frac{4(s+4)}{s+2}},$$

and multiplying numerator and denominator by $(s+1)(s+2)$,

$$T(s) = \frac{8 + 2(s+4)(s+1)}{4(s+4)(s+1)} = \frac{2s^2 + 10s + 16}{4(s+1)(s+4)} = \frac{s^2 + 5s + 8}{2(s+1)(s+4)}.$$

Reading the answer. The closed-loop poles are at -1 and -4 , and the d.c. gain is $T(0) = 8/8 = 1$. The interesting part is the numerator: the feedforward path has given the system a pair of *zeros*, at $s = -2.5 \pm j1.32$, where the chain alone had none. This is Week 3's lesson arriving from a new direction — a parallel route through a system puts zeros in its transfer function, and those zeros reshape the transient without changing a single pole.

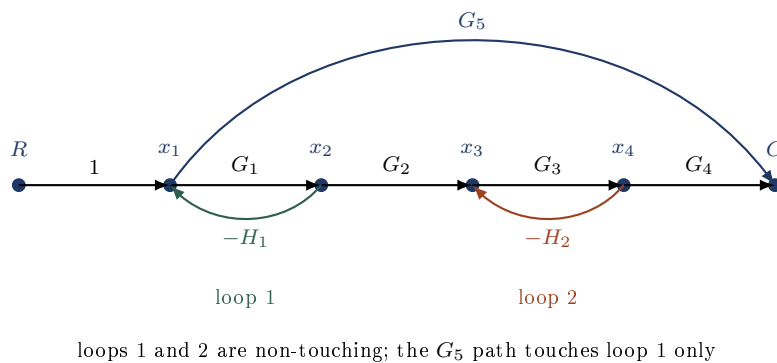


Figure 8: Worked example 4.5: a graph with two forward paths (the chain, and the feedforward branch G_5) and a non-touching pair of loops. Cf. Nise §5.5, p. 251.

8 Systems with more than one input

An LTI system's response to several inputs acting at once is the sum of its responses to each acting alone — superposition, which holds precisely because the system is linear. In block-diagram terms: to find the transfer function from one input, set all the other inputs to zero and

reduce the diagram as usual; then add the contributions.

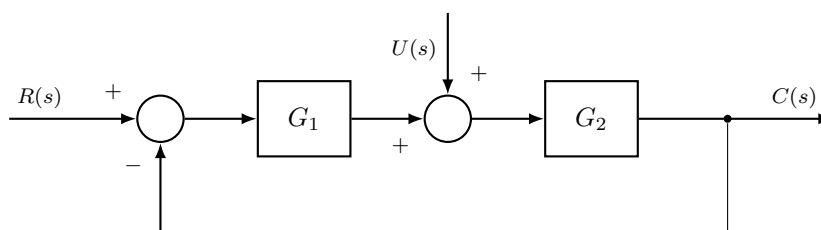


Figure 9: A system with a second input U entering between the two forward blocks. Its response is found by superposition: one reduction with $U = 0$, a second with $R = 0$.

Take Figure 9 with $G_1 = 2$, $G_2 = 3/(s + 1)$ and unity feedback. Setting $U = 0$ leaves the canonical form with forward path $G_1G_2 = 6/(s + 1)$:

$$\frac{C(s)}{R(s)} = \frac{G_1G_2}{1 + G_1G_2} = \frac{\frac{6}{s+1}}{1 + \frac{6}{s+1}} = \frac{6}{s+7}.$$

Setting $R = 0$ instead, U sees only G_2 on the way out, but the same loop on the way round:

$$\frac{C(s)}{U(s)} = \frac{G_2}{1 + G_1G_2} = \frac{\frac{3}{s+1}}{1 + \frac{6}{s+1}} = \frac{3}{s+7},$$

and the total response is

$$C(s) = \frac{6}{s+7}R(s) + \frac{3}{s+7}U(s).$$

The two transfer functions share a denominator — necessarily, since both are governed by the same characteristic equation $1 + G_1G_2 = 0$ — and differ only in numerator, because U enters the loop further downstream and so is amplified less. Here a unit step at U produces half the steady-state output of a unit step at R .

Key idea

Superposition means one characteristic equation, several numerators. Extra inputs change *what* the system responds to; they cannot change *how* it responds, because the poles belong to the loop, not to the input.

Where this stops in FEE3411. When the second input is something unwanted — a load torque, a supply fluctuation, sensor noise — it is called a disturbance, and the interesting question becomes how to design a loop that rejects it. Computing the response, as above, is analysis and belongs here. Designing for disturbance rejection, and dealing with plant uncertainty and measurement noise, is FEE3412's work; Khalil §4-4 to §4-8 and Nise §7.5 and §7.7 are worth reading for interest, but are not examinable in this unit.

Summary

- A block diagram has three elements — block, summing junction, pick-off point — and encodes one algebraic equation in s per block and per junction. A pick-off point copies a signal; it does not divide it.

- Cascade gives G_1G_2 , parallel gives $G_1 \pm G_2$, negative feedback gives $G/(1 + GH)$. The cascade rule assumes no loading between stages.
- In canonical form, G is the forward path, GH the open-loop transfer function, $T = G/(1 + GH)$ the closed-loop transfer function and $1/(1 + GH)$ the error transfer function. The closed-loop poles are the roots of the characteristic equation $1 + GH = 0$.
- Feedback does not change the number of poles; it relocates them. That relocation is the whole purpose of closing a loop.
- Block-diagram algebra adds two moving rules: a summing junction moved against the flow divides by the block, and a pick-off point moved with the flow needs a compensating $1/G$ on the moved branch.
- Nested loops reduce from the inside out. *Overlapping* loops do not — they need a junction move first, which is where hand reduction becomes error-prone.
- A signal flow graph replaces blocks with branch gains and summing junctions with nodes, absorbing feedback signs into the branch gains.
- Mason's gain formula, $T = \sum_k P_k \Delta_k / \Delta$, reads the transfer function off the graph with no redrawing. Δ alternates signs over loop gains, then products of non-touching pairs, then triples; Δ_k deletes the loops that touch path k . "Non-touching" means sharing no node.
- Several inputs are handled by superposition: one transfer function per input, all sharing the same denominator.

Before the tutorial

1. A unity-feedback system has $G(s) = 12/[s(s + 8)]$. Write down its open-loop transfer function, its characteristic equation, and its closed-loop transfer function. Are the closed-loop poles real or complex?
2. In the canonical form, which of G , GH and $G/(1 + GH)$ would a question mean by "the loop gain"? Which sets the closed-loop poles?
3. A pick-off point sits before a block $G(s) = 4/(s + 3)$ and feeds a branch of gain 2. Redraw it with the pick-off taken *after* the block, and state the new branch gain.
4. Draw the signal flow graph of a unity-feedback system whose forward path is a cascade of G_1 and G_2 , then use Mason's formula to confirm you get $G_1G_2/(1 + G_1G_2)$. How many forward paths and loops did you find?

Looking ahead

This week finished the modelling half of FEE3411. From Week 2's differential equations, through Week 3's transfer functions, to this week's interconnection rules, we can now write down a single $T(s)$ for a complete control system — and every remaining week of the unit is about interrogating that one function. Week 5 starts immediately: it takes the ω_n and ζ read off denominators like $s^2 + 3s + 4$ in Worked example 4.2 and turns them into overshoot, rise time, peak time and settling time. Week 6 asks what steady-state error the response settles to, and Week 7 asks the question this week quietly set up — whether the roots of $1 + G(s)H(s) = 0$ are all in the left half plane at all. Assignment 2, issued this week and due in Week 6, covers Weeks 3 and 4 together: transfer functions, block-diagram reduction, signal flow graphs and Mason's rule. The boundary holds as always — locating a closed-loop pole is *analysis* and belongs here; choosing $G(s)$ to put it somewhere better is *design*, and belongs to FEE3412.